FinTrust | Planning and design

Release 1 | Planning and design evidence | 13 October 2026

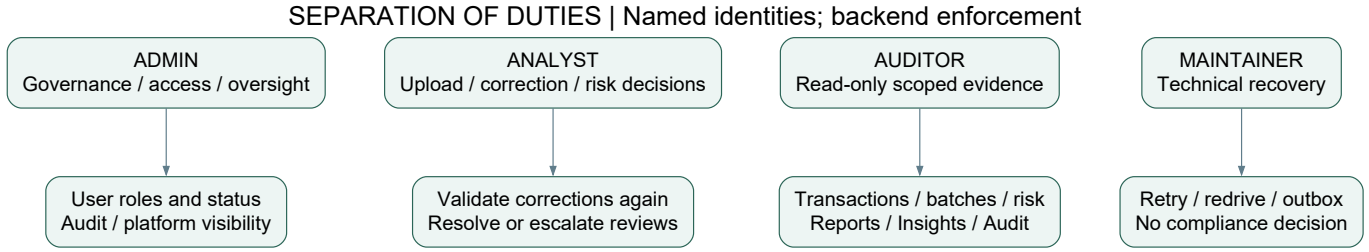
1. Project problem

Transaction exports need reliable admission, duplicate protection, explainable risk signals and accountable human review. FinTrust records and investigates financial events. It does not move money, reverse charges or establish fraud.

2. Users and roles

Role	Accountability
Administrator	Governance, users/access, role assignment, disable/reactivate and oversight.
Risk / Compliance Analyst	CSV upload, quarantine correction, investigation and review decisions.
Auditor	Read-only scoped records, reports and audit history.
Technical Maintainer	Technical failure diagnosis, controlled retry/redrive and recovery.

Technical recovery != compliance decision. Admin visibility != routine Analyst authority.



Evidence boundary

The frozen local application uses isolated PostgreSQL and local service substitutes. AWS stages 1-4 are recorded as verified; historical ingest remains disabled. Live schema bootstrap and complete application wiring remain pending. This document creates no new AWS evidence.

Admission | Files and rows

3. Core use case

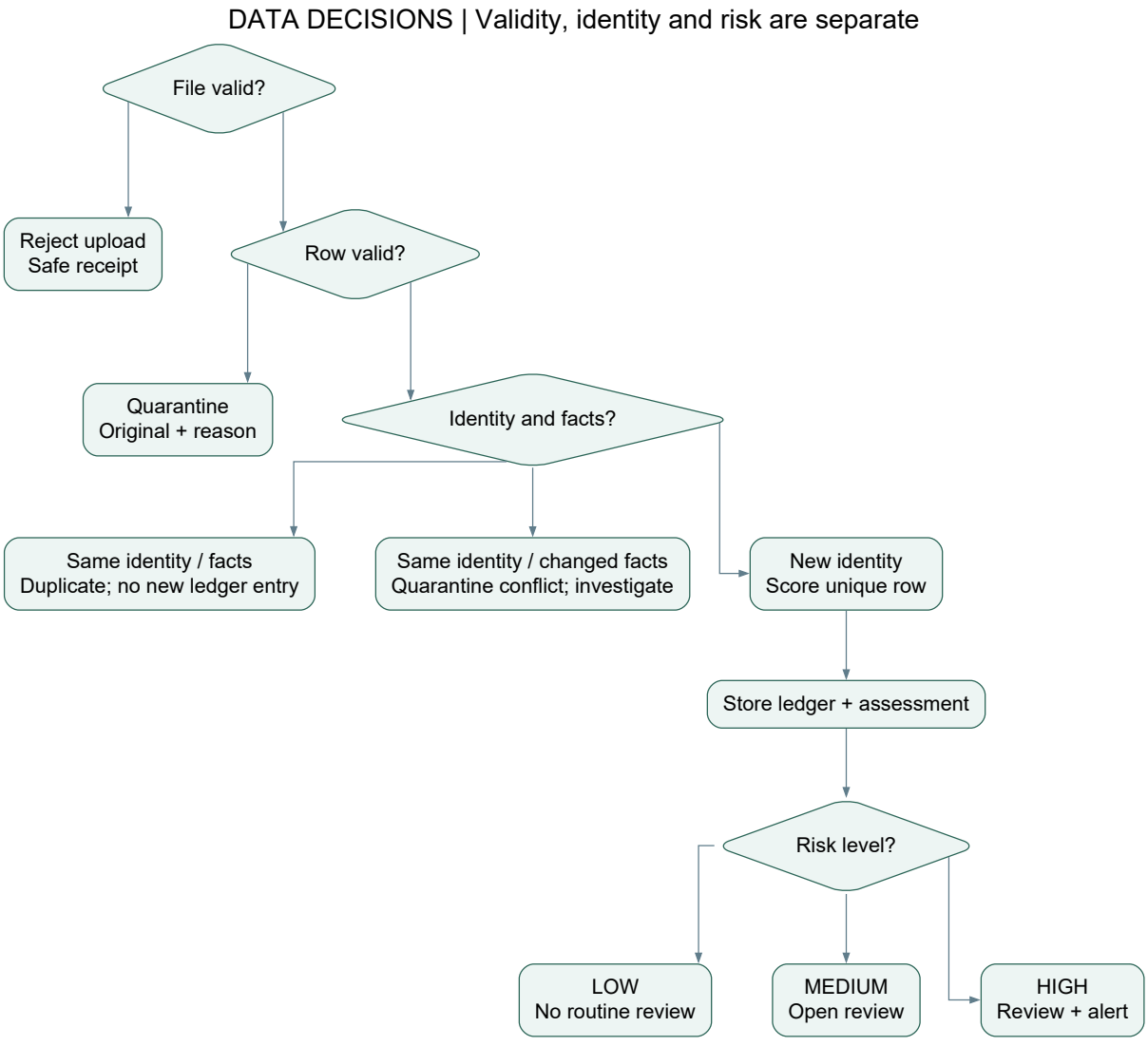
An Analyst admits a scoped CSV, reviews its reconciled outcomes, corrects invalid business rows and investigates risk cases. An Auditor can inspect the resulting evidence without changing it.

4. File-ingestion rules

CSV only; maximum 1 MiB and 5,000 rows. Invalid structure or unsupported files produce a safe rejection receipt and no ledger processing. Exact file replay returns the original batch rather than processing a second time.

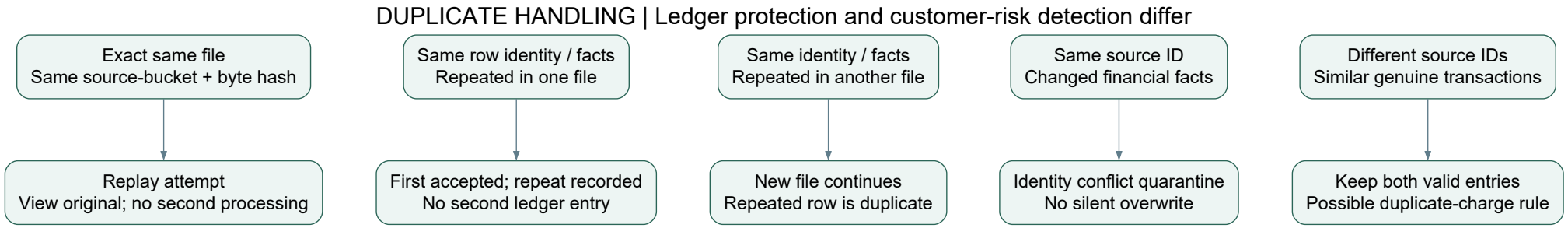
5. Row-validation rules

Require source transaction identity, account, valid timestamp and positive amount. Validate supported currency, country and transaction type; scoring confirms the account exists. Missing/unknown account, malformed or future timestamp, invalid amount and zero/negative amount are quarantined. Negative rows are not silently interpreted as refunds.



Identity | Duplicate decisions

6. Duplicate-handling decisions



Case	Release 1 outcome
Exact same file hash in the source bucket	Audit replay attempt; no second processing; view original batch.
Same transaction in the same file / across files	Record occurrence; no second ledger entry. A new file still processes its other rows.
Same source identity with changed financial facts	Quarantine identity conflict; investigate; no silent overwrite.
Different trusted IDs with similar genuine facts	Keep both accepted events; raise possible duplicate-charge risk if conditions match.

Duplicate protection protects the ledger. Duplicate-charge detection protects the customer.

Quarantine | Risk and review

7. Quarantine and correction

Preserve safe original payload, reason code/detail, source file, batch, row and actor. Correction records a new occurrence/version; the worker revalidates it and repeats identity checks. The original is retained. Rejected/invalid data is distinct from technical failure work.

8. Frozen risk-scoring rules

Component	Points / boundary
Amount risk	0 / 15 / 25 / 40; strongest band only.
High velocity	25; more than 5 accepted in the closed 300-second window, current included.
Cross-border	15; country differs from ZA / explicit cross-border flag.
Possible duplicate	20; distinct source references match debit/merchant/amount/currency within inclusive +/-120 seconds.

V2: fintrust-r2-rules-2026-10-13. Amount <=50,000: 0; >50,000 to 100,000: 15; >100,000 to 200,000: 25; >200,000: 40. Thresholds use transaction currency, without FX. V1 scores and versions remain frozen.

9. Review workflow

Band	Outcome
LOW: 0-19	Store ledger and assessment; no routine review.
MEDIUM: 20-39	Store and open review.
HIGH: 40-100	Store, open review and enter the independent alert path.

OPEN -> UNDER_REVIEW -> CLEARED / CONFIRMED_RISK / ESCALATED. Backend-enforced transitions preserve named actor, notes, versions and history. A risk flag is a reason to investigate, not proven fraud.

Explainability | Measured risk components

Controlled local example: 65% - HIGH

Transaction -> Measured values -> Components -> Score -> Risk level -> Analyst investigation. This is deterministic rule scoring, not machine learning.

Measurement	Component / contribution
150,000 ZAR	Amount band >100,000 and <=200,000: +25.
8 accepted entries, 09:55-10:00 UTC	More than 5 in the closed 300-second window: high velocity +25.
Transaction country US; baseline ZA	Cross-border +15.
No matching distinct charge	Possible duplicate +0.
Total	65% - HIGH. Investigate and choose a human review outcome.

Rule-based risk score. This percentage represents the configured risk score, not the probability of fraud.

Evaluation timestamp, configured thresholds, observed windows, references and rule version are frozen in the original SQL result. Historical V1 evidence is displayed without reconstructing missing measurements. See docs/EVIDENCE/12-e2e/risk-model-v2/presentation-example.json for the real isolated test result.

Migration 006 changes only the existing review-eligibility guard: V1 LOW 30 stays ineligible; V2 MEDIUM 25 can open review. All accepted LOW/MEDIUM/HIGH transactions remain in the ledger.

If I had more time and budget | Planned

Behaviour-aware monitoring - not implemented

Unusual amount versus account baseline; unusual time and after-hours activity; unfamiliar country/geography; unusual channel; new counterparty; dormant-account reactivation; bursts; rapid movement of funds; repeated payments below thresholds; historical behaviour deviation.

A transaction at night is not automatically suspicious. It becomes a stronger signal when it differs from the account's normal behaviour. More history and context, testing and governance are required before these dimensions affect a score.

Production hardening - not capacity proof

Load tests at 10k / 100k / 1M transactions; Lambda concurrency and connection tuning; FIFO message-group strategy; high-throughput FIFO if justified; RDS Proxy; queue-depth alarms; Multi-AZ; source-feed monitoring; configurable SLAs; grouped alerts; disaster-recovery tests.

Advanced intelligence - future scope

Streaming ingestion, anomaly detection, an AI Investigation Assistant using authorised evidence and human review, richer customer/account context and governed adaptive thresholds. No AI or ML was added to Release 1.

Current architecture versus proven scale

Designed: SQS buffers bursts; event-driven Lambda workers use account ordering and controlled concurrency; RDS owns ledger/workflow, DynamoDB owns projections/claims; technical retries/DLQ; no NAT and cost-conscious limits. Live wiring remains pending.

The architecture is designed for elastic processing, but production-scale performance testing and connection tuning remain a future hardening step. A design is not a claim of tested throughput.

Keep the Analyst in view: What happened? Why is it risky? What should I do next? Future features must improve one of those answers.

Use cases | Separation of duties

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Scope follows the signed-in identity and assigned accounts. Sign out before changing roles; do not reuse another person's session.



10. Failure handling / DLQ

Quarantine concerns business data; retry/dead-letter queues concern technical processing. FIFO groups work by account. Durable correction/projection outboxes preserve pending intent after partial failures. Retry must not duplicate ledger entries or alerts; uncertain email outcomes need reconciliation.

Received rows	Must resolve to visible outcomes
Accepted	One canonical ledger identity.
Quarantined	Original and reason retained; controlled correction.
Duplicate	Occurrence retained; no new ledger entry.
Pending / technical failure	Visible and recoverable; nothing disappears silently.

11. Audit and accountability

Keep named human subjects, timestamps, reason/notes, version history and operation receipts. Local identities use an ignored file-backed registry with hashed tokens; governance receipts use the S3-style receipt store. Do not invent SQL users/roles or notifications tables. Browser notices are session-local.

12. Reports and Insights

Scoped reports and Insights expose actual financial and workflow aggregates, batch outcomes, review queues and account history. CSV/JSON exports support evidence; summaries are not a complete restorable archive. No ledger deletion is approved by a summary report.

Presentation references such as ACC-0005, TX-0180, BATCH-1042 and RISK-0091 are readable examples, not replacement identity keys. UUIDs remain authoritative where the implementation uses them.

Database | Meaning and relationships

13. Database / ERD

Term	Meaning
File	Uploaded CSV.
Batch	Processing record for an accepted file.
Row	One financial-event candidate with source row lineage.
Account	Financial account being monitored.
Transaction	Accepted financial event in the ledger.
Quarantine	Invalid/conflicting business row awaiting a controlled decision.
Risk event	Frozen scoring evidence for visibility and, by score band, review.

File -> Batch -> Rows -> Transactions. Account -> Transactions. Transaction -> Risk event -> Review. These are process relationships; only declared catalog foreign keys are drawn as physical relationships.

The verified local PostgreSQL 16.15 catalog contains 17 tables and two derived views: batch_reconciliation and quarantine_workflow. Transactions.batch_id is lineage, not a declared foreign key. Source-system/account/source-transaction identity is conditionally unique; fingerprints are not globally unique.

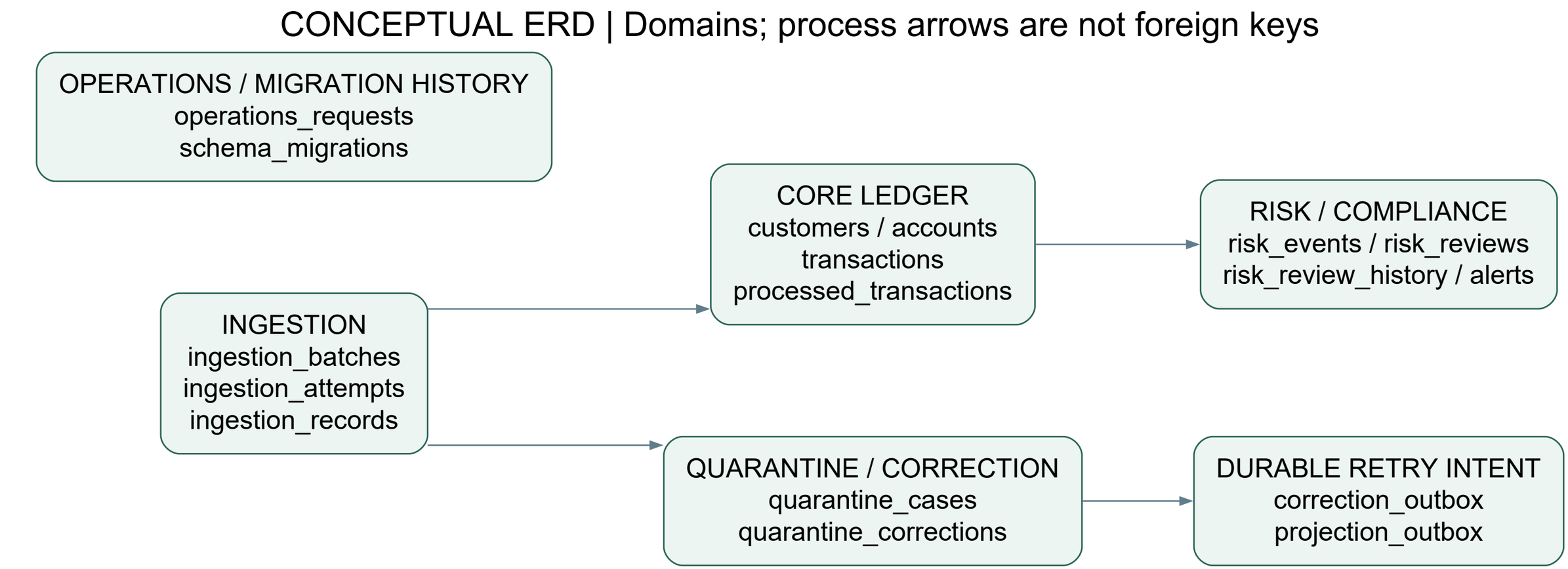
S3 holds original/safe payloads and reports; DynamoDB holds immutable projections and alert claims. SQL owns ledger, workflows, outboxes and migration history. Live RDS schema bootstrap remains pending.

The following conceptual and four domain pages preserve readable physical keys and cardinalities. PK/FK/UK markers describe primary/foreign/unique keys; composite markers apply together. The retained schema notes provide exact constraint definitions.

FinTrust | Conceptual domain overview

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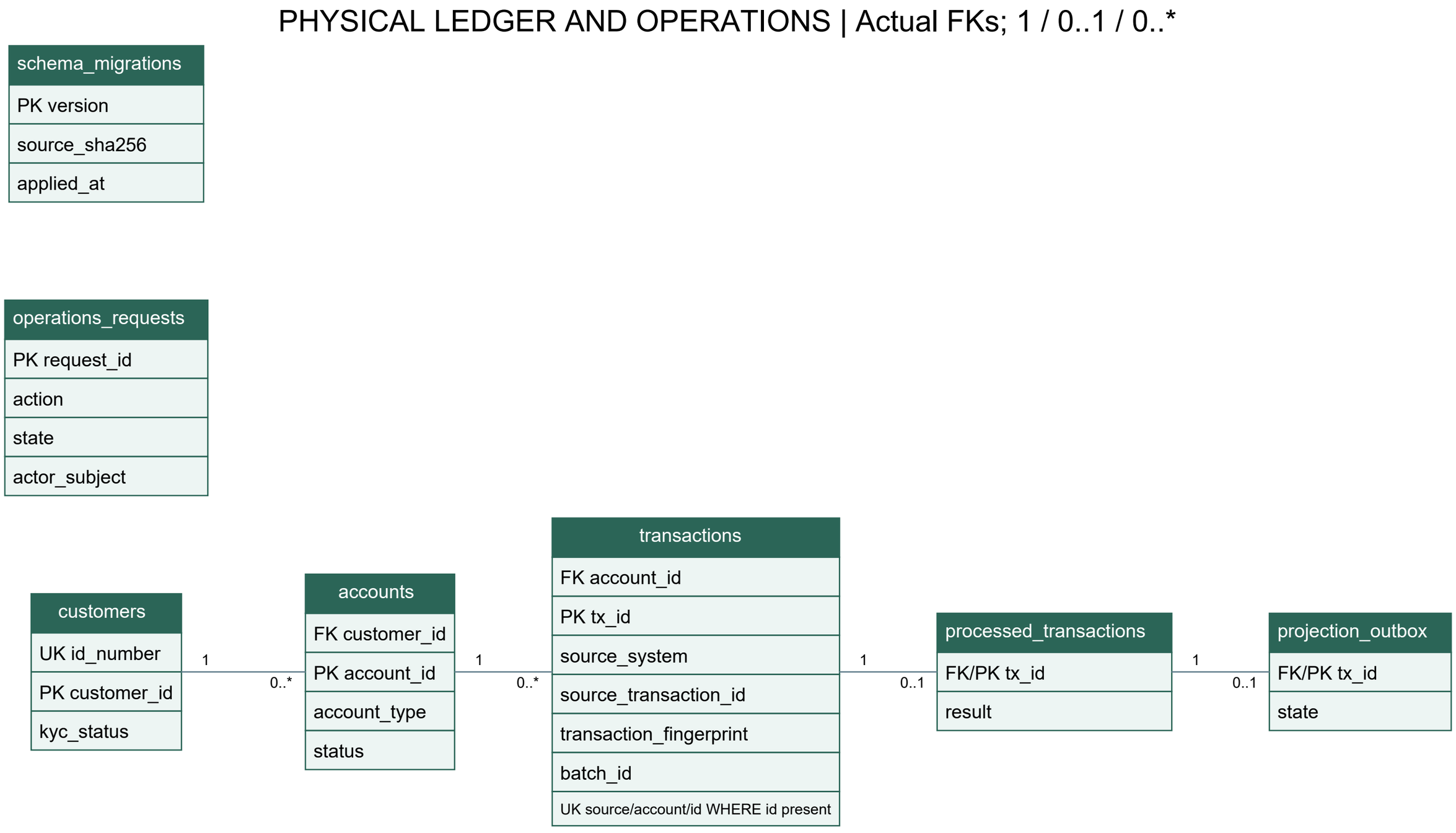
Conceptual domain map: process arrows, not foreign keys.



FinTrust | Physical ledger and operations

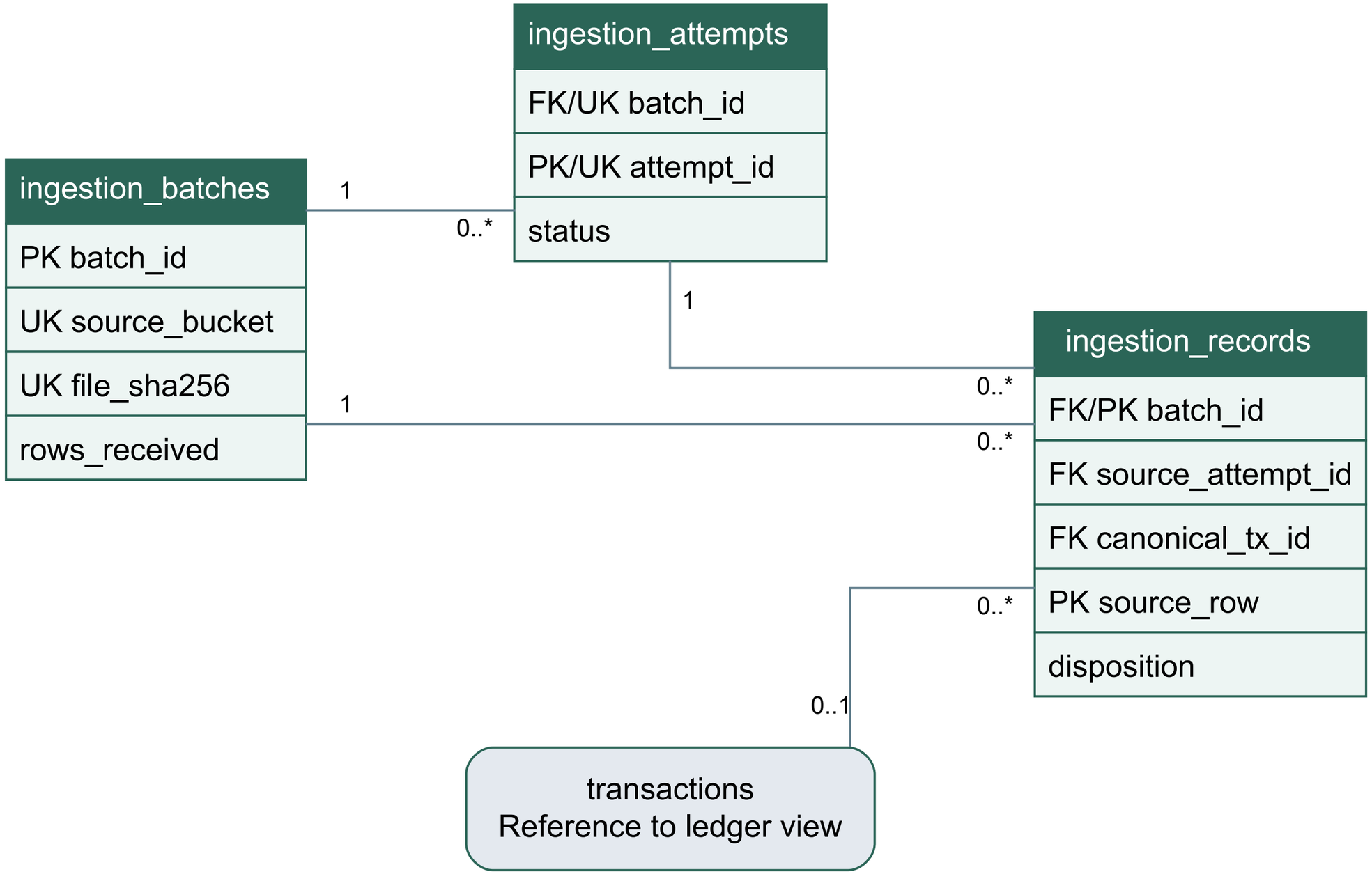
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Physical foreign keys and keys are unchanged. Grey reference boxes connect domains across pages. PK / FK / UK: primary / foreign / unique keys. All 17 tables and 16 foreign keys are preserved.



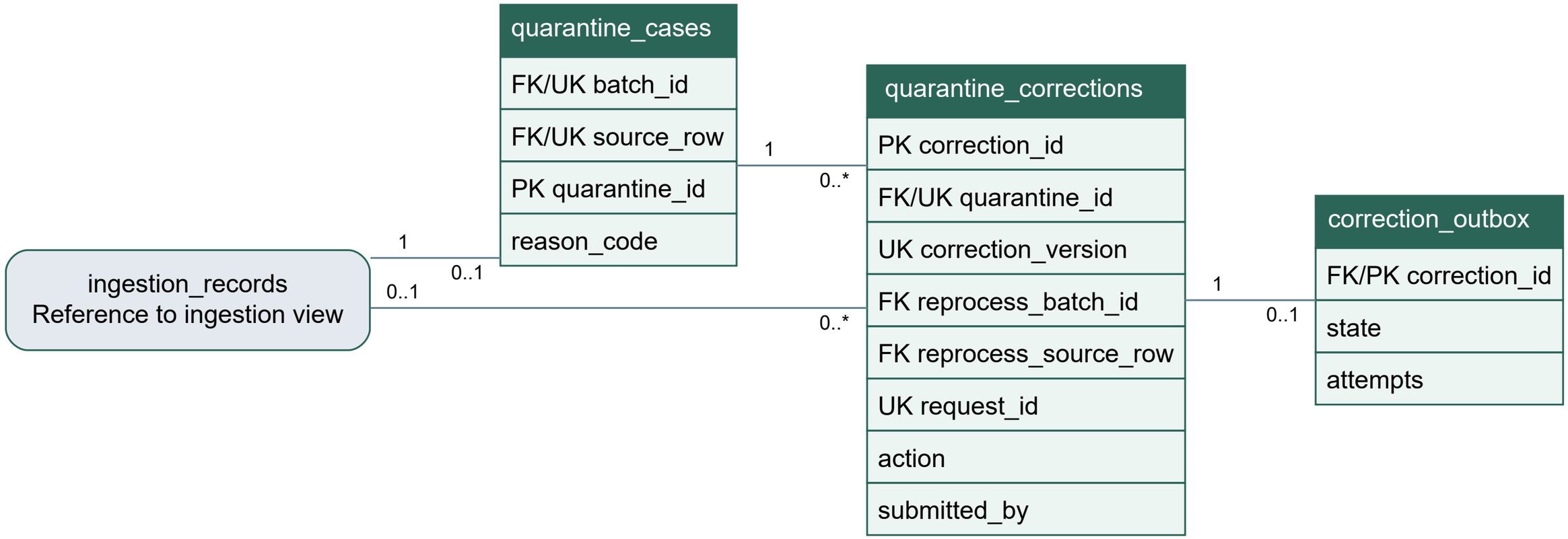
Physical foreign keys and keys are unchanged. Grey reference boxes connect domains across pages. PK / FK / UK: primary / foreign / unique keys. All 17 tables and 16 foreign keys are preserved.

PHYSICAL INGESTION | Actual FKs; 1 / 0..1 / 0..*



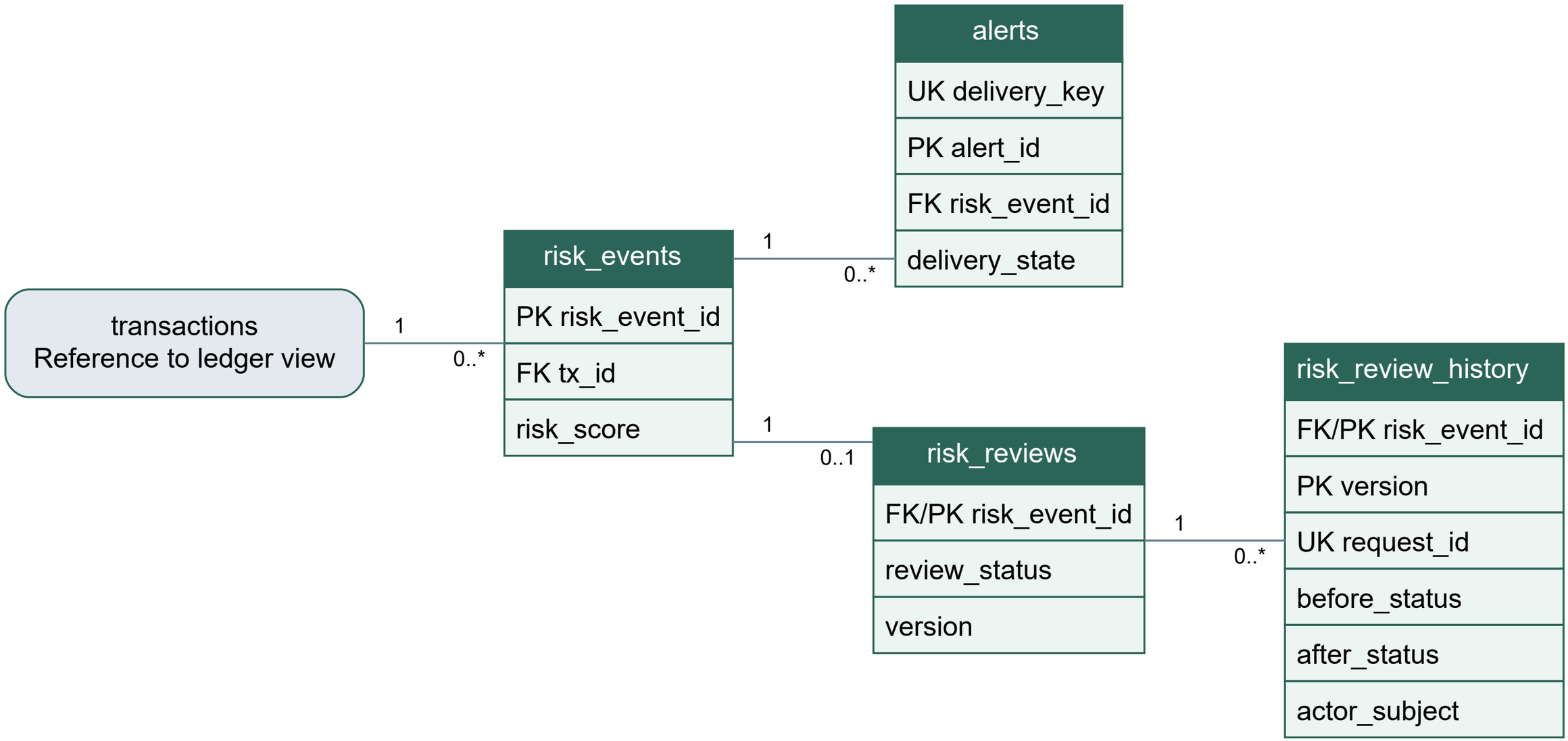
Physical foreign keys and keys are unchanged. Grey reference boxes connect domains across pages. PK / FK / UK: primary / foreign / unique keys. All 17 tables and 16 foreign keys are preserved.

PHYSICAL QUARANTINE AND CORRECTIONS | Actual FKs; 1 / 0..1 / 0..*



Physical foreign keys and keys are unchanged. Grey reference boxes connect domains across pages. PK / FK / UK: primary / foreign / unique keys. All 17 tables and 16 foreign keys are preserved.

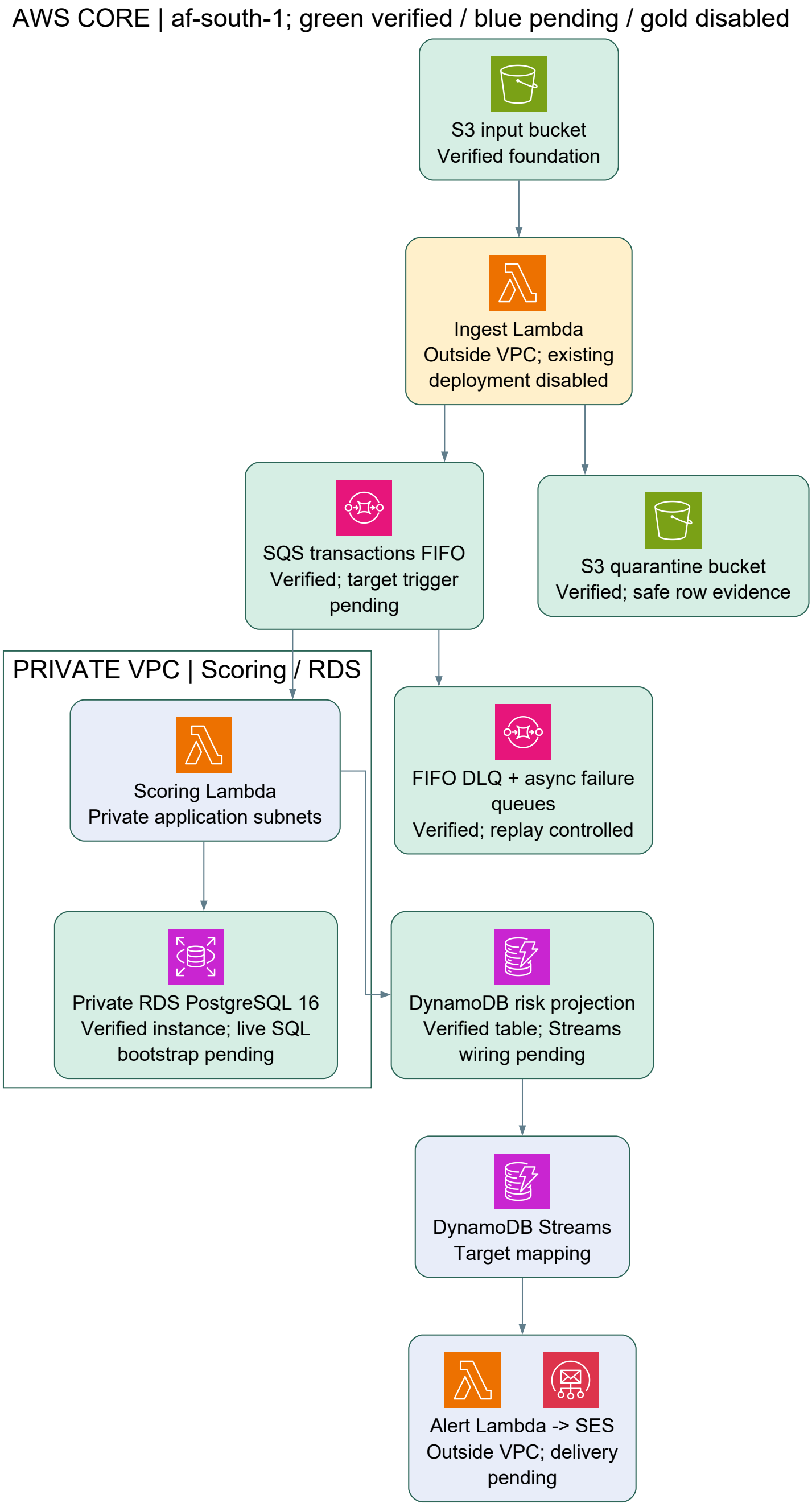
PHYSICAL RISK AND REVIEW | Actual FKs; 1 / 0..1 / 0..*



FinTrust | Core processing and failures

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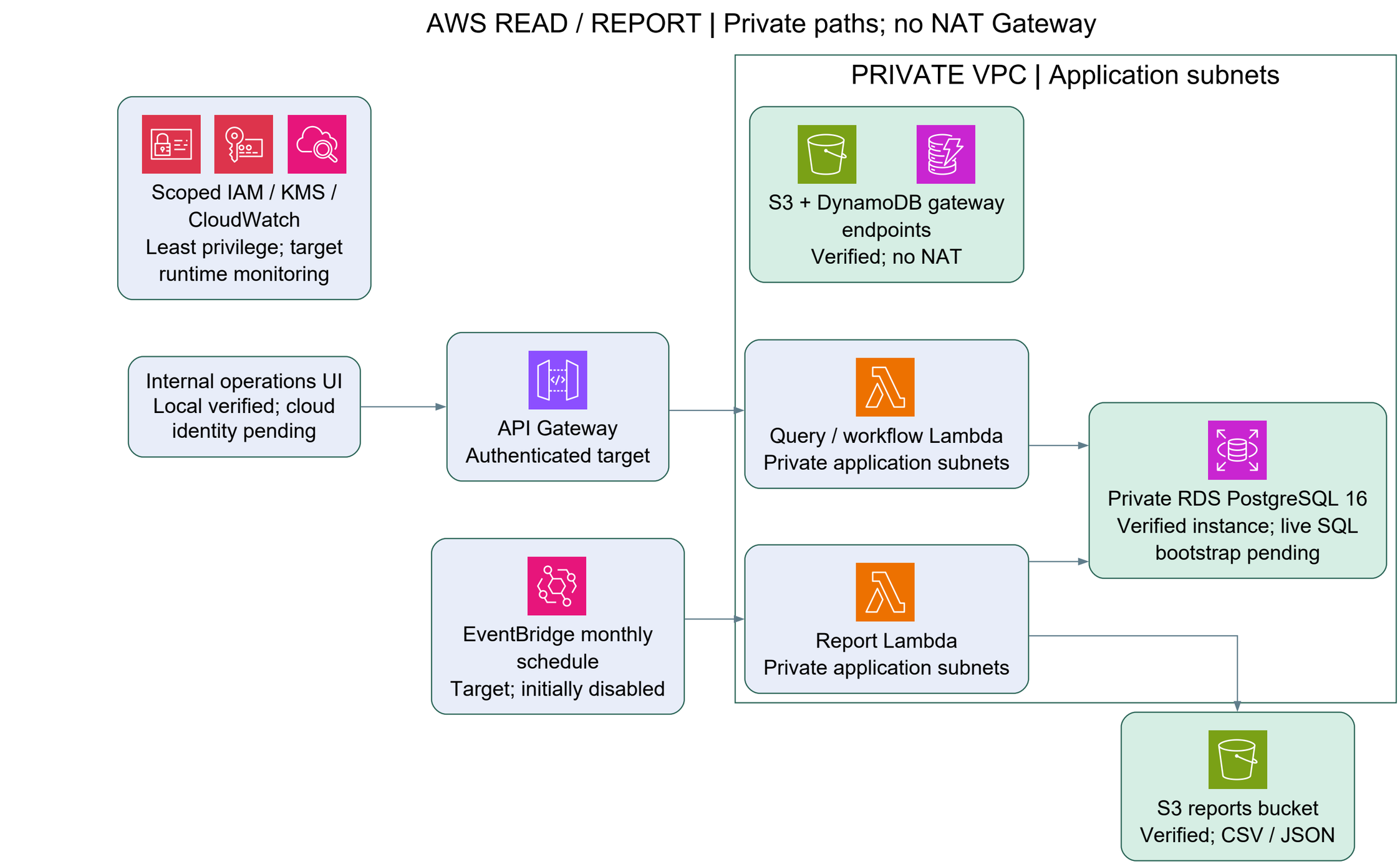
Official AWS service icons are embedded. GREEN: verified foundation; BLUE: Release 1 target with pending wiring; GOLD: disabled historical ingest. Outline: private VPC. Region af-south-1; no NAT Gateway. Icons do not prove deployment.



FinTrust | Query reporting and security

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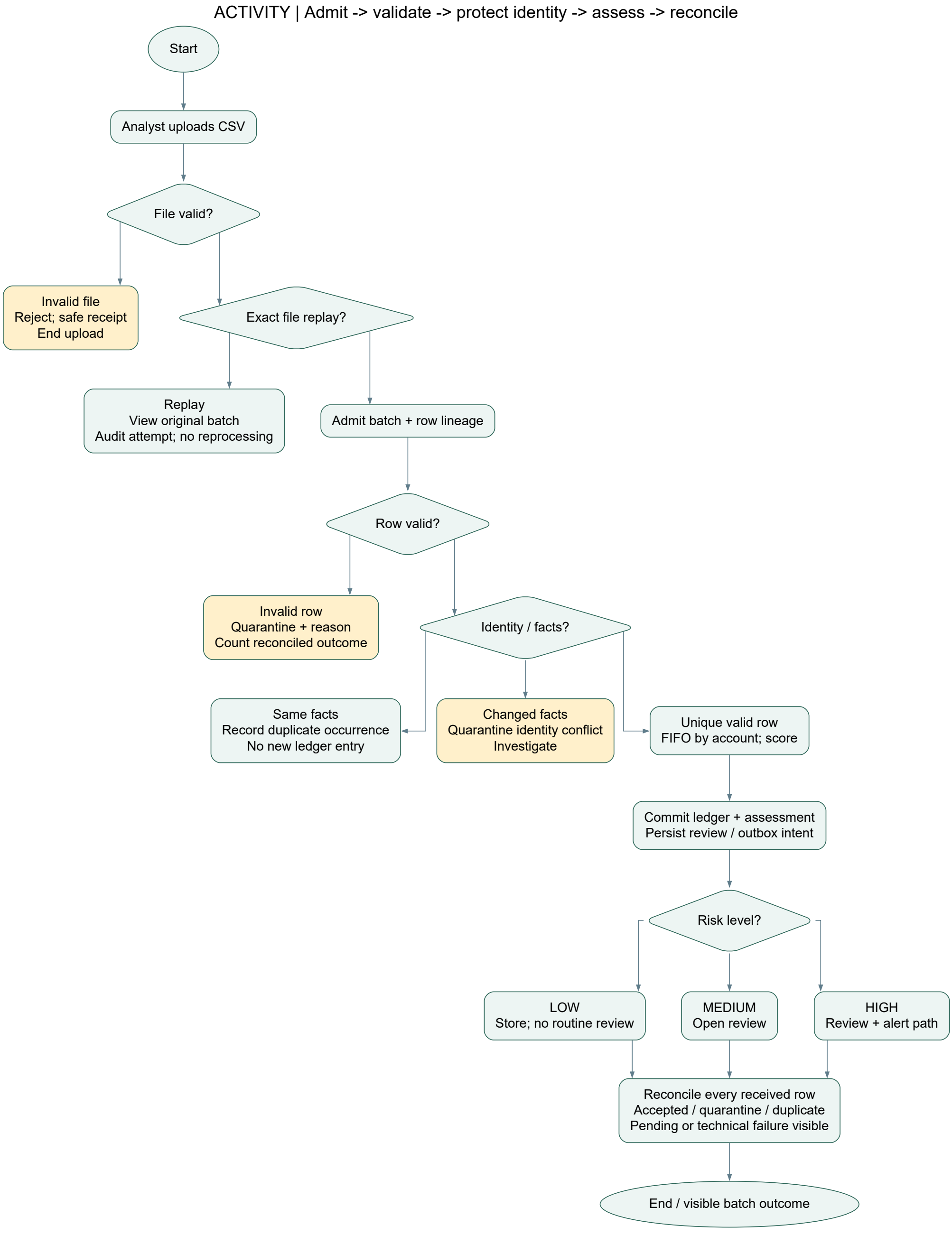
Official AWS service icons are embedded. GREEN: verified foundation; BLUE: Release 1 target with pending wiring; GOLD: disabled historical ingest. Outline: private VPC. Region af-south-1; no NAT Gateway. Icons do not prove deployment.



FinTrust | Admission and reconciliation

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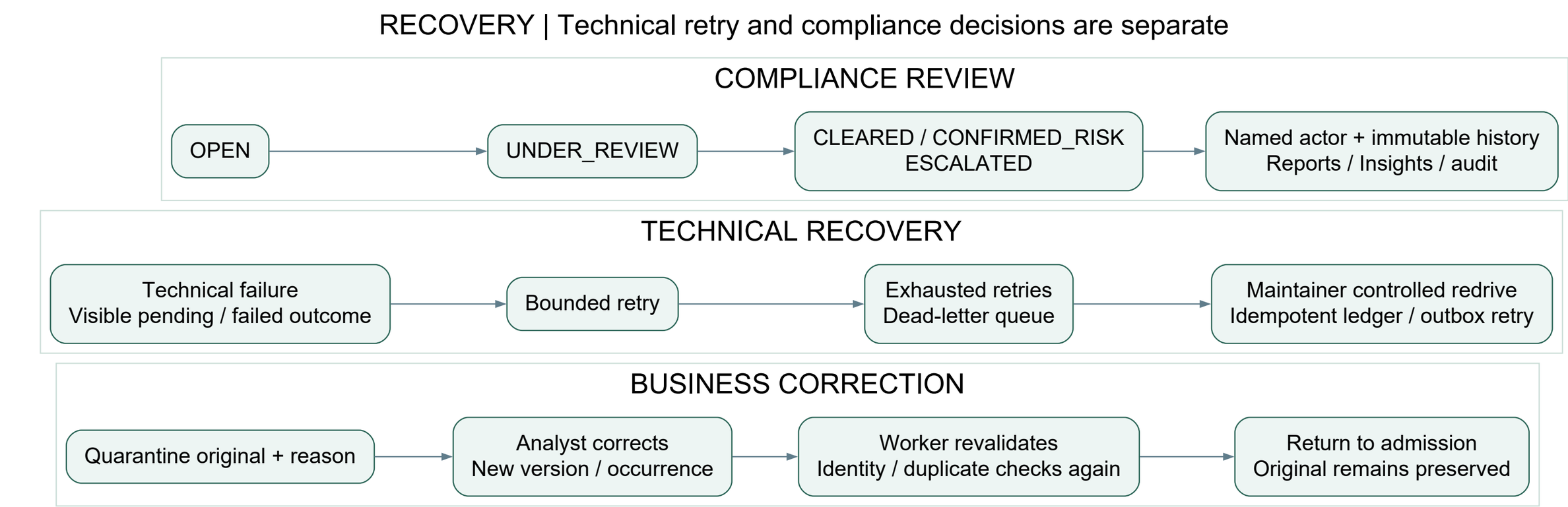
Read admission and scoring from top to bottom. Separate recovery and review paths are on the second page. Invalid files are rejected; invalid rows are quarantined. A controlled technical retry is not a compliance decision.



FinTrust | Correction recovery and review

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Read admission and scoring from top to bottom. Separate recovery and review paths are on the second page. Invalid files are rejected; invalid rows are quarantined. A controlled technical retry is not a compliance decision.



Release boundary | Present safely

17. Release 1 scope vs deferred work

- Locally verified: scoped roles, bounded CSV admission, identity protection, quarantine/correction, frozen risk review, reconciliation, reports/Insights and accountability.

- Recorded AWS foundation: private network, S3/KMS, private PostgreSQL instance, SQS/DynamoDB; historical ingest disabled.

- Pending approval and verification: live base schema/migration chain, hardened final Linux candidate, cloud identity/API, triggers, report schedule, SES delivery and complete live end-to-end wiring.

- Deferred: velocity changes to hour-based rules, automatic refunds/reversals, proven fraud classification and complete archive/deletion workflow.

Short live-demo script

- Use the existing Monday runbook for local launch and approved fixtures; never seed or migrate live RDS for presentation.

- Sign in as the prepared Analyst. Show assigned scope and reconciled accepted / duplicate / quarantine outcomes.

- Show file rejection versus row quarantine, lineage and an approved correction fixture.

- Show LOW / MEDIUM / HIGH reasons and review history, then reports/Insights and audit.

- Sign out, then demonstrate Auditor read-only scope. Explain Admin and Maintainer duties with the role diagram.

- Close with the AWS diagram. Say explicitly which components remain pending; do not improvise technical retries or claim mocked email was delivered.

CSV A/B were created outside the repository and are not assumed to exist at invented paths. If a fixture is unavailable, show verified evidence and state that limitation. Keep credentials out of slides, logs and evidence.

References | Source and maintenance

Authoritative sources

- Current risk-model baseline: a474566 plus the reviewed V2 refinement. Prior diagram checkpoint c6f4274 remains architectural design evidence; its icons do not prove live wiring.

- Verified local catalog: docs/EVIDENCE/05-rds/presentation-ux-freeze/postgres16-integrity.json.

- Canonical schema detail: docs/diagrams/fintrust-release1-erd-notes.md. Business contract: docs/RELEASE1_BUSINESS_PROCESS.md.

- Current local readiness: docs/RELEASE1_PRESENTATION_READINESS.md. Launch/fixture procedure: docs/RELEASE1_MONDAY_DEMO.md.

- Official AWS icon attribution: aws.amazon.com/architecture/icons/. Icons are embedded unchanged from the retained service assets; no AWS deployment is implied.

Reproduction and file discipline

Seven DOT sources under docs/presentation/diagram-source/ produce intermediate SVGs under ignored build/presentation-pdf/svg/. tools/render_release1_diagrams.cjs performs the vector layout and embeds official icons. tools/build_presentation_pdf.py produces the eight final PDFs. Requires Node/Viz.js, Python/reportlab/svglib/PyMuPDF and Arial fonts.

Current planning Markdown and current standalone SVG duplicates are removed only after PDF validation. Canonical development Markdown, historical screenshots, deployment/recovery history, AWS icon assets and project-reference remain retained. No AWS, application, migration or push operation is part of conversion.

Historical design inventory

Archive-only exports retained under docs/EVIDENCE/12-e2e/historical-design-diagrams/: architecture_day1_day3.svg, architecture_target.svg, erd.svg, activity.svg, use_cases.svg, upload_sequence.svg, fintrust.drawio, fintrust_architecture_and_erd.pdf and fintrust_system_documentation.pdf. Their hash/move manifest remains in the diagram-pack evidence folder.

No history is discarded for visual appearance. The two historical Python renderers write only into that archive. Current authority is one planning PDF, seven diagram PDFs and seven editable DOT sources. Current-file references are updated to the PDF set; no extra current PDF versions are kept.

Inventory | Final files and support notes

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PDF	Purpose / editable source
FinTrust_ERD.pdf	Database / ERD / fintrust-release1-erd.dot
FinTrust_AWS_Architecture.pdf	AWS architecture / fintrust-release1-aws-architecture.dot
FinTrust_Activity_Diagram.pdf	End-to-end activity / fintrust-release1-activity.dot
FinTrust_Use_Case_Diagram.pdf	Use cases / fintrust-release1-use-cases.dot
FinTrust_Data_Decision_Flow.pdf	Data decisions / fintrust-release1-data-decisions.dot
FinTrust_Duplicate_Handling.pdf	Duplicate handling / fintrust-release1-duplicate-handling.dot
FinTrust_Role_Separation.pdf	Role separation / fintrust-release1-role-separation.dot
FinTrust_Project_Planning_and_Design_Evidence.pdf	Combined planning, rules, demo and inventory / tools/build_presentation_pdf.py

Project-reference: keep unchanged today

File	Classification / recommendation after presentation
README.md / HANDOFF.md	Keep useful navigation/handoff; review against current readiness.
presentation_notes.md / live_demo_script.md	Keep safety net; use the current PDF script as authority; consolidate later.
learning_guide.md	Keep learning/reference history.
completed_work.md / component_guide.md	Duplicate or dated milestone narrative; consolidate after comparison.
local_software_setup.md	Earlier setup duplicate; consolidate into canonical Operations after review.
daily_log.md	Preserve chronological history; archive/consolidate later.
requirements_day1_day3.md / upcoming_features.md	Outdated milestone/future scope; archive or consolidate after presentation.

No project-reference file is approved for deletion today. SAFE TO DELETE AFTER PRESENTATION applies only to copies confirmed redundant in a later comparison. Six earlier inline diagram duplicates already point to the authoritative views.